



Cardiac: Wide Complex Tachycardia with Pulse

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA Tape.
- Focus on rapid and early BLS airway and ventilation tools. Intubation may not be the best option for these patients.
- Pediatric pads should be used in children < 10 Kg or PEDIA tape color purple.

Signs & Symptoms:

- QRS > 0.12 sec
- Ventricular tachycardia on ECG (runs or sustained)
- Conscious, rapid pulse
- Chest pain
- Shortness of breath
- Dizziness
- Rate usually 150-180 bpm for sustained V-tach

Differential:

- Artifact / Device failure
- Cardiac history
- Endocrine / Electrolyte
- Hyperkalemia
- Drugs / Toxic exposure
- Pulmonary disease
- Tricyclic OD

Clinical Management Options

P	P	P	P	P	P		• Oxygen PRN titrated to SpO₂ 92%-96% • Basic airway management
L	L	L	L	L	L		
1	2	3	4	5	6		• 4 lead and 12 lead ECG placement and acquisition • Apply waveform EtCO₂
							• Vascular access • Isotonic Crystalloid PRN titrated to SBP ≥ 90 mmHg or MAP ≥ 65
							• Monitor and interpret ECG • If a Treatable cause is identified, move that treatment up in priority • If Torsades de Pointes: Magnesium Sulfate • If Torsades de Pointes: Defibrillate at maximum joules for Adult • If Ventricular Tachycardia: Amiodarone infusion • If VT is refractory Amiodarone, then Lidocaine • Sedate with Midazolam or Ketamine • If hyperkalemia or Tricyclic OD, consider Sodium Bicarbonate early • Synchronize cardioversion at maximum Joules for Adult • For Pediatric Cardioversion 0.5 – 1.0 j/kg, repeat at 2 j/kg as needed • 12 lead ECG post conversion

Consult Online Medical Control As Needed

PEDIATRIC CARIOVERSION DOSE CHART

Preemie	Newborn	4MO	6MO	1YR	2YR	3YR	4YR	5YR	6YR	7YR	8YR	9YR	10YR	11YR	12YR	13YR
2KG	4KG	6KG	8KG	10KG	12KG	15KG	17KG	20KG	22KG	25KG	27KG	30KG	35KG	40KG	50KG	60KG
1j	2j	3j	4j	5j	6j	8j	9j	10j	10j	15j	15j	15j	20j	20j	30j	30j
2j	4j	6j	8j	10j	10j	15j	15j	20j	20j	30j	30j	30j	30j	50j	50j	70j
4j	8j	10j	15j	20j	20j	30j	30j	50j	50j	50j	50j	70j	70j	85j	100j	120j



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Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- For witnessed / monitor ventricular tachycardia, try having patient cough while preparing other therapies.
- Slow wide complex, consider Hyperkalemia.
- Maximum dose of antiarrhythmic should be given before changing antiarrhythmic.
- Amiodarone: allow 10 minutes after each dose completed before next dose.
- Pediatric pads should be used in children < 10 kg or PEDIA tape color purple.
- Consider a change of vector if initial Cardioversion is unsuccessful to anterior/posterior pad placement.
- Sinus tachycardia may be misinterpreted as SVT or A-Fib. Sinus tachycardia rate > 150 bpm in the adult patient or > 180 in the pediatric patient may be seen in the septic patient.